Weather Driven Business Intelligence Presentation Script

Three Presenter Version

Document purpose	A coordinated 12 to 15 minute script with speaking lines handoffs demonstration cues and question preparation
Prepared	12 October 2026
Presentation date	13 October 2026

Weather driven decisions explained with transparent assumptions and reproducible analysis

How to use this script

Replace Presenter 1 Presenter 2 and Presenter 3 with your names. Rehearse once with the dashboard open and once without it. The script targets approximately 13 minutes and leaves time for questions. If the live weather service is unavailable use the fallback data and explain that the application is designed to remain demonstrable during an API outage.

Presenter	Primary responsibility	Target time
Presenter 1	Problem project overview and data engineering	4 minutes
Presenter 2	Econometric method scenario logic and interpretation	4 minutes
Presenter 3	Dashboard demonstration evaluation and conclusion	4 to 5 minutes

Presenter 1 Opening and data foundation

Cue Slide 1 Title

Good morning. Our project is called the Weather Driven Business Intelligence Platform. It explores a practical business question: when weather changes customer behavior and operating conditions how can a company estimate the likely impact and respond before the disruption becomes expensive?

Different businesses react in opposite ways to the same event. Heavy rain may increase food delivery and rideshare demand while reducing restaurant footfall bike usage and outdoor retail activity. Our platform brings those effects into one portfolio view instead of examining each business in isolation.

Cue Slide 2 Project objective

The platform has three objectives. First it aligns weather and business records that arrive at different frequencies and in different timezones. Second it estimates weather relationships after accounting for normal calendar patterns. Third it translates the analysis into scenarios charts and operational guidance that a nontechnical manager can understand.

An important point is that the historical dataset is synthetic. We designed it to demonstrate a realistic analytical workflow across delivery mobility retail hospitality energy and freight. We do not present the simulated results as verified effects from real companies.

Cue Slide 3 Data sources and scale

The first raw source contains 69 106 hourly weather observations. The second contains 279 249 business records recorded every 15 minutes. The data cover New York City Chicago Seattle and Phoenix during 2023 and 2024.

The sources deliberately do not line up. Weather timestamps are in UTC while business timestamps are local. Weather is hourly while business activity is every 15 minutes. Some weather readings are missing. Solving these mismatches is a central part of the project rather than an afterthought.

Cue Slide 4 Data pipeline

The pipeline assigns each city its correct IANA timezone converts local business records to UTC aggregates business activity to hourly intervals and joins both sources by city and hour. It then creates calendar indicators rain categories weather lags temperature anomalies and cross sector ratios.

The final analytical warehouse contains 70 045 synchronized hourly observations and 61 variables. In our latest verification it contained no missing values and no duplicate city hour keys. This clean warehouse becomes the input to the modeling layer.

Cue Handoff to Presenter 2

With the data foundation established Presenter 2 will explain how we separate weather relationships from recurring business patterns and how the scenario engine uses those estimates.

Presenter 2 Method and business interpretation

Cue Slide 5 Why correlation can mislead

A raw correlation is not enough. Suppose delivery orders are high on a rainy Friday evening. Rain may contribute but Friday evenings are normally busy even in clear weather. If we ignore that calendar pattern we can attribute too much of the increase to rain.

For each of 21 business metrics the project fits an ordinary least squares regression containing precipitation snow temperature anomaly and wind. It also controls for hour of day day of week month holiday status and city. Robust standard errors are used to reduce sensitivity to unequal residual variance.

Cue Slide 6 Interpretation and boundaries

The result is a calendar and city adjusted estimate. For example a rain coefficient describes the modeled change associated with one additional millimeter of precipitation while the included controls remain fixed. We compare this adjusted estimate with the raw correlation to show how confounding can change an apparent relationship.

We deliberately use the phrase adjusted association rather than proven causal effect. The data are simulated and the model does not create a natural experiment. R squared describes in sample model fit and a p value is evidence relative to a null hypothesis. Neither should be described as guaranteed predictive accuracy or certainty.

Cue Slide 7 Cross sector scenario

The scenario engine lets a user change city weekday hour rain snow temperature anomaly and wind. It compares a clear weather baseline with the selected conditions across six sectors.

Consider a heavy Friday evening storm. Bike usage may fall as commuters move toward transit and rideshare. More vehicles can increase congestion. Restaurant dine in demand may decline while delivery demand rises. At the same time courier availability may fall which can extend delivery times and increase cancellations. This is the domino effect the dashboard is designed to communicate.

These chains are plausible scenario narratives rather than a separately estimated causal network. Their purpose is to help managers consider second order effects such as staffing packaging capacity courier incentives inventory positioning and freight delays.

Cue Handoff to Presenter 3

Presenter 3 will now show how these ideas appear in the dashboard and summarize the current strengths limitations and next steps.

Presenter 3 Dashboard demonstration and conclusion

Cue Slide 8 or live dashboard Portfolio controls

The application is built with Streamlit and Plotly. At the top we select one of four cities and choose a portfolio from more than 16 preset business categories. We can also create a custom business by specifying its baseline revenue and weather sensitivities.

Cue Demonstration Show Weekly Operational Impact

The first tab compares the modeled revenue series with a clear weather baseline and displays the net weather related shift. The historical tab expands the time horizon and includes the raw versus adjusted comparison discussed earlier.

Cue Demonstration Show Domino Chains

The interactions tab shows how a weather event may move demand between businesses and create downstream pressure. The descriptions change with the selected city businesses and weather regime.

Cue Demonstration Show Live Scenario Simulator

The live scenario tab uses Open Meteo for current and seven day weather conditions. Revenue projections then apply weekday patterns and configured business sensitivities. The guidance is rule based and transparent. It is not a black box AI prediction.

NASA POWER is used as a recent reference source and NASA Worldview provides an external satellite imagery link. Because the NASA reference can be delayed it should not be interpreted as a same moment confirmation of current conditions.

Cue Slide 9 Verification and limitations

The project includes seven automated checks covering data generation alignment scenario output connector schemas CSV detection and elasticity estimation. All seven tests passed in our latest run.

The main limitation is that the default business history and the automatic Square and Stripe streams are simulated. The connector classes contain live API paths but the application does not provide credentials by default. A merchant can upload CSV or Excel history but those estimates still require enough overlapping dates and proper validation.

Cue Slide 10 Conclusion

In conclusion this project demonstrates a complete analytical pipeline from mismatched raw feeds to business focused decision support. Its strengths are modular engineering explainable assumptions cross sector thinking and an accessible interface.

The next step is to validate the approach with permissioned real merchant data report genuine uncertainty and backtest the forecasts. That would move the platform from a strong portfolio prototype toward a dependable business intelligence product. Thank you. We are happy to take questions.

Fast version if time is reduced

For an eight minute slot each presenter should keep the opening and closing paragraph of their section plus one technical paragraph. Skip the detailed NASA explanation and show only the portfolio selector scenario tab and diagnostics tab during the demonstration.

Likely questions and suggested answers

Question	Suggested answer
Is the data real	The historical cross sector dataset is synthetic. Current weather can be live and users may upload real merchant revenue for exploratory calibration.
Why not use simple correlation	Weather and commerce share calendar and seasonal patterns. The adjusted regression controls for several of those recurring patterns before estimating the weather relationship.
Does the model prove causality	No. It estimates adjusted associations. Causal claims would require stronger identification and validation on observed data.
Where is AI used	The present forecast and advice layers are deterministic and rule based. We avoid claiming a trained AI model where none is implemented.
Are Square and Stripe live	The classes support credential based calls but the default application runs calibrated sandbox sequences. Production integration needs secure credentials pagination and monitoring.
How would you improve accuracy	Use real merchant histories add holdout backtesting uncertainty intervals clustered or panel appropriate inference and forecast benchmarks.
What happens if an API fails	The weather module returns a marked fallback package so the demonstration remains usable. The UI should always show whether data are live or fallback.

Rehearsal checklist

- Replace presenter labels with names and agree on pronunciation of econometrics and heteroskedasticity.
- Open the dashboard before presenting and confirm the city selector charts and scenario controls respond.
- Keep a screenshot or PDF of the main views available in case network access fails.
- Do not call the synthetic revenue real or the adjusted regression causal.
- Pause briefly at each handoff and let the next presenter begin before changing slides.
- Aim for one complete rehearsal tonight and one timed rehearsal tomorrow before presenting.